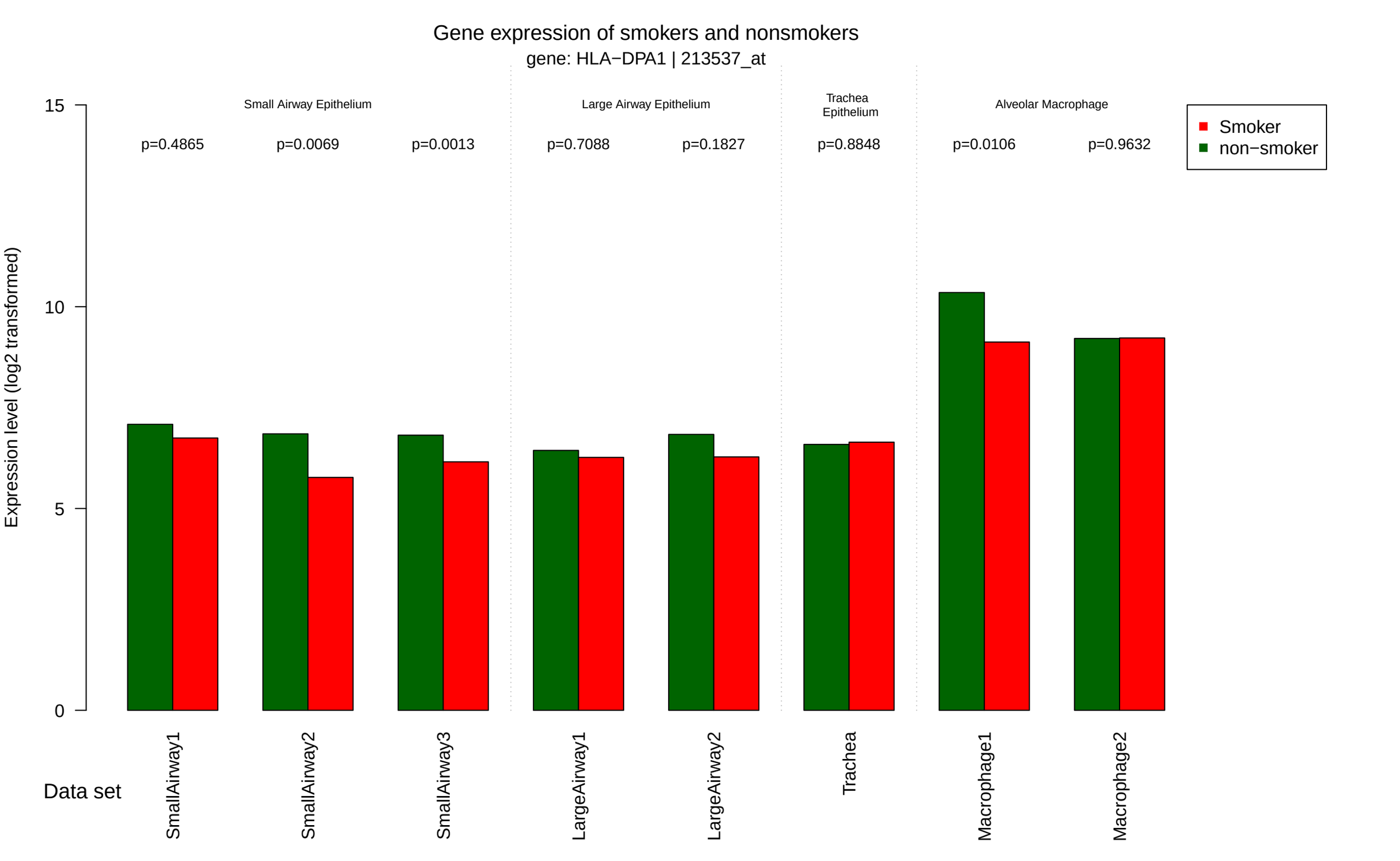
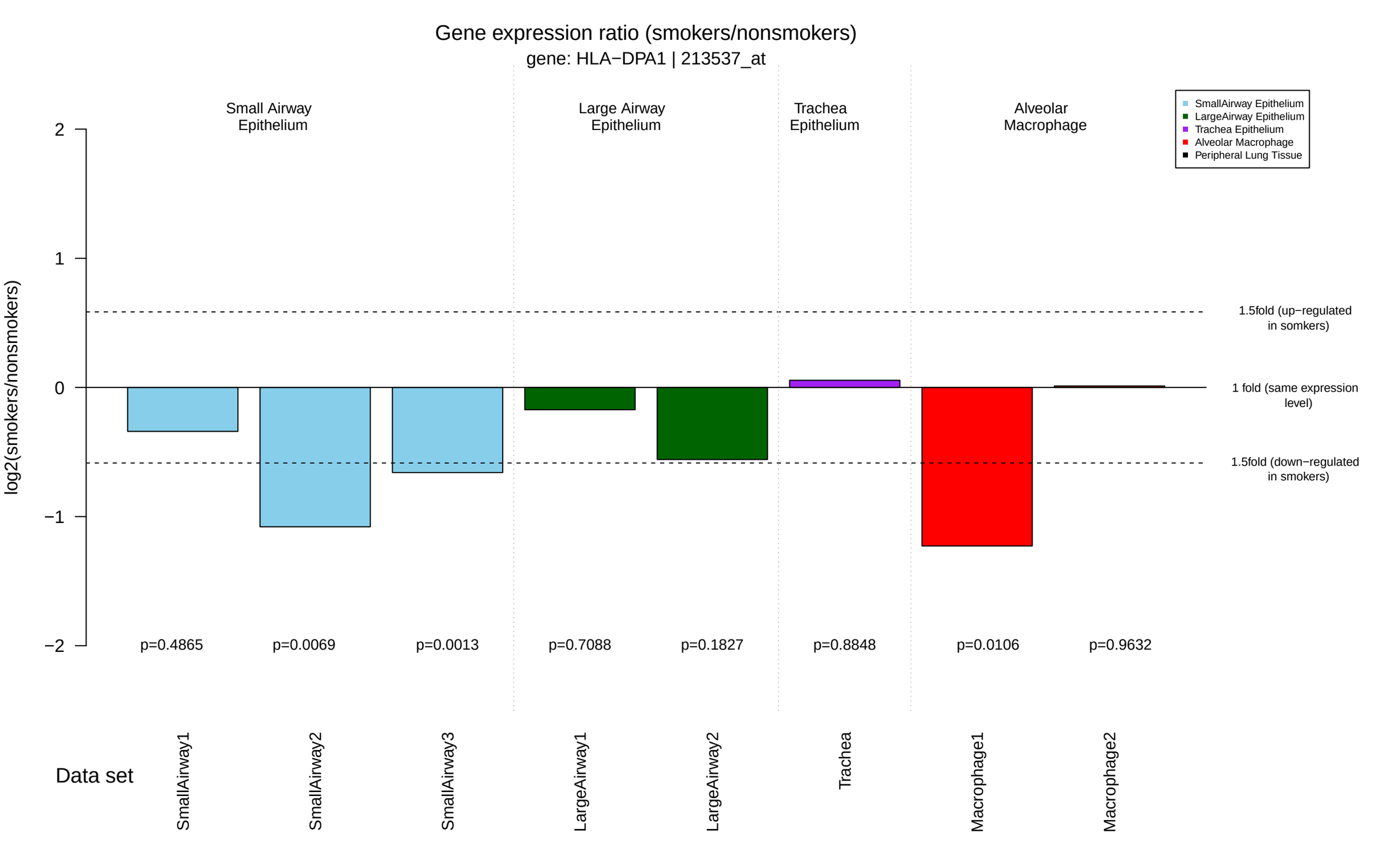
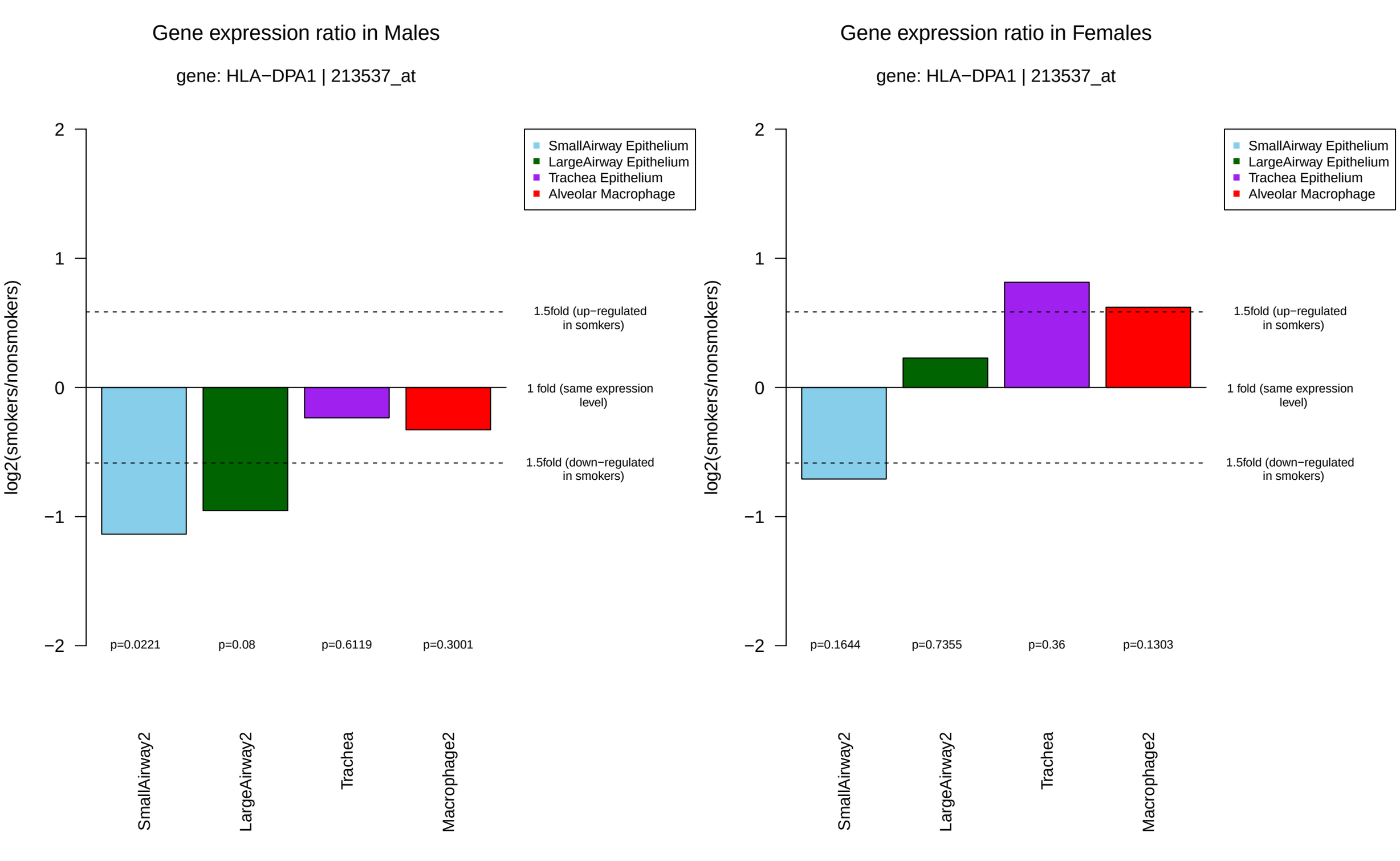
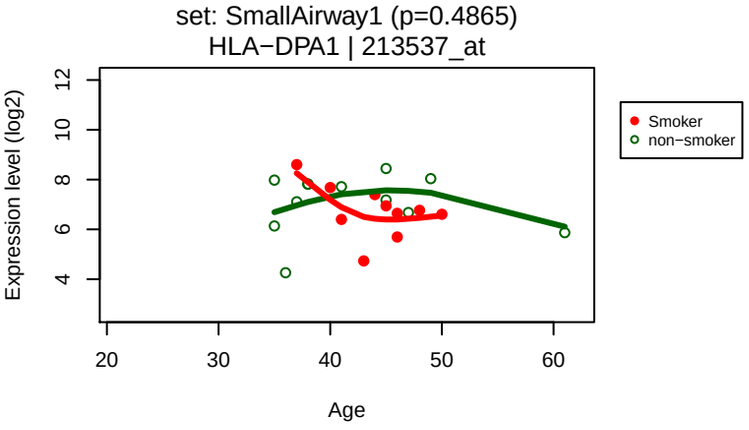
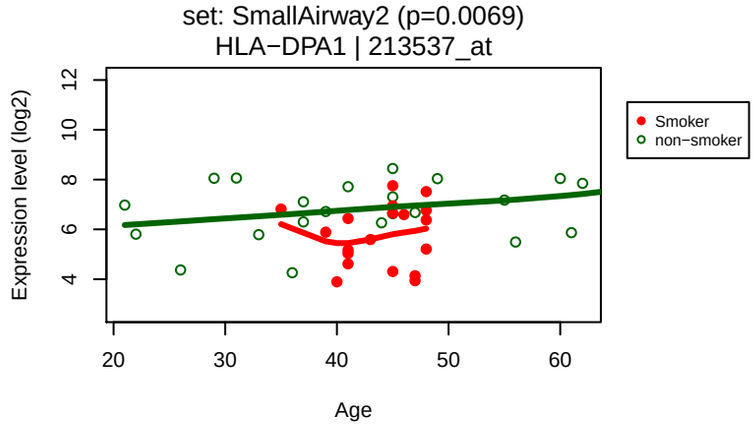
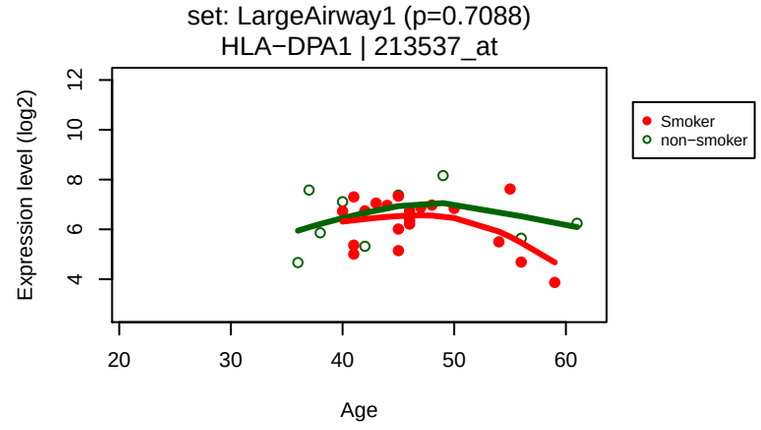

Gene expression of smokers and nonsmokers
gene: HLA−DPA1 | 213537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4865
p=0.0069
p=0.0013
p=0.7088
p=0.1827
p=0.8848
p=0.0106
p=0.9632
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

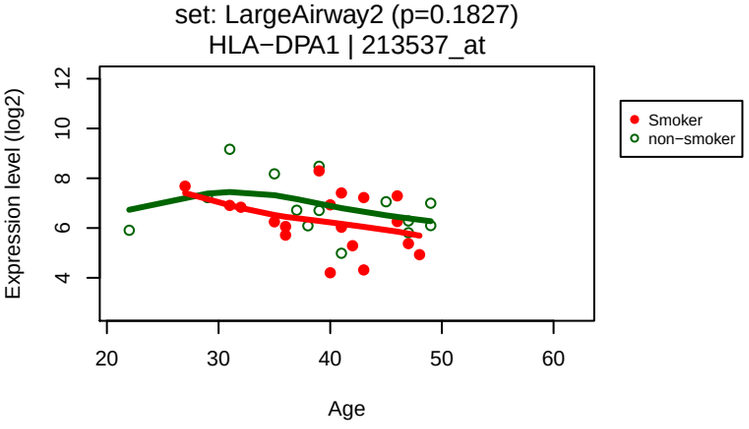

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DPA1 | 213537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4865
p=0.0069
p=0.0013
p=0.7088
p=0.1827
p=0.8848
p=0.0106
p=0.9632
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

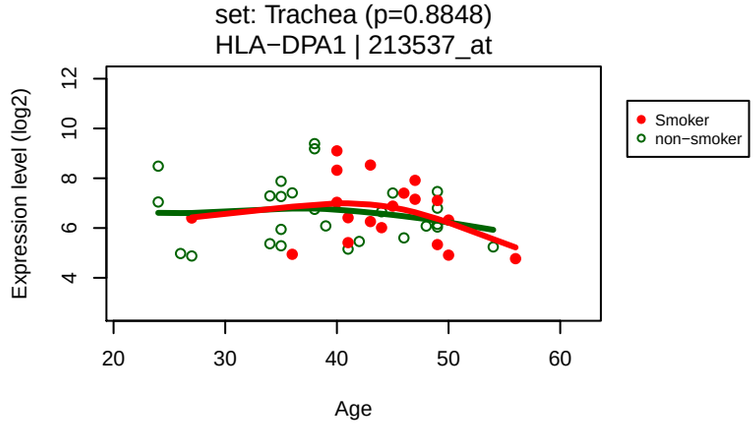

Gene expression ratio in Males
gene: HLA−DPA1 | 213537_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0221
p=0.08
p=0.6119
p=0.3001
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DPA1 | 213537_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1644
p=0.7355
p=0.36
p=0.1303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

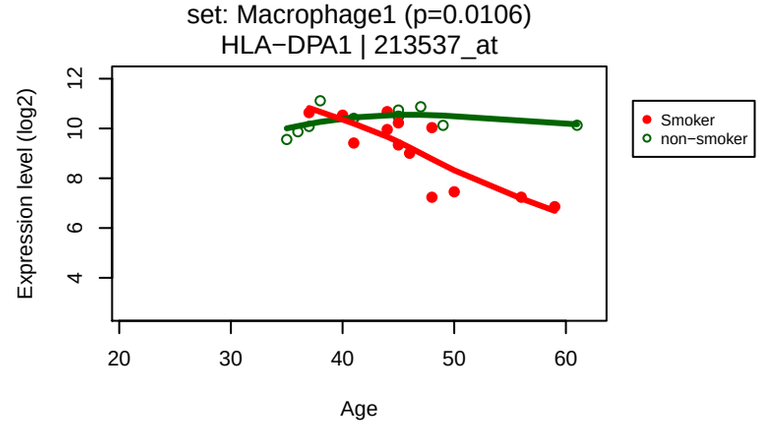

set: SmallAirway1 (p=0.4865)
HLA−DPA1 | 213537_at
Expression level (log2)
Age
Smoker
non−smoker

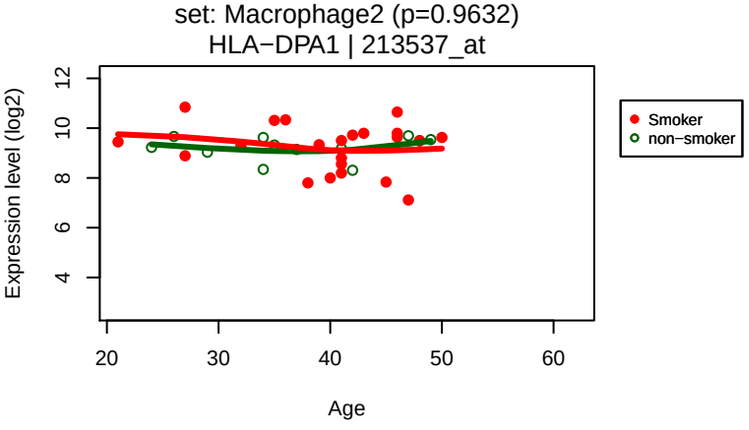

set: SmallAirway2 (p=0.0069)
HLA−DPA1 | 213537_at
Expression level (log2)
Age
Smoker
non−smoker

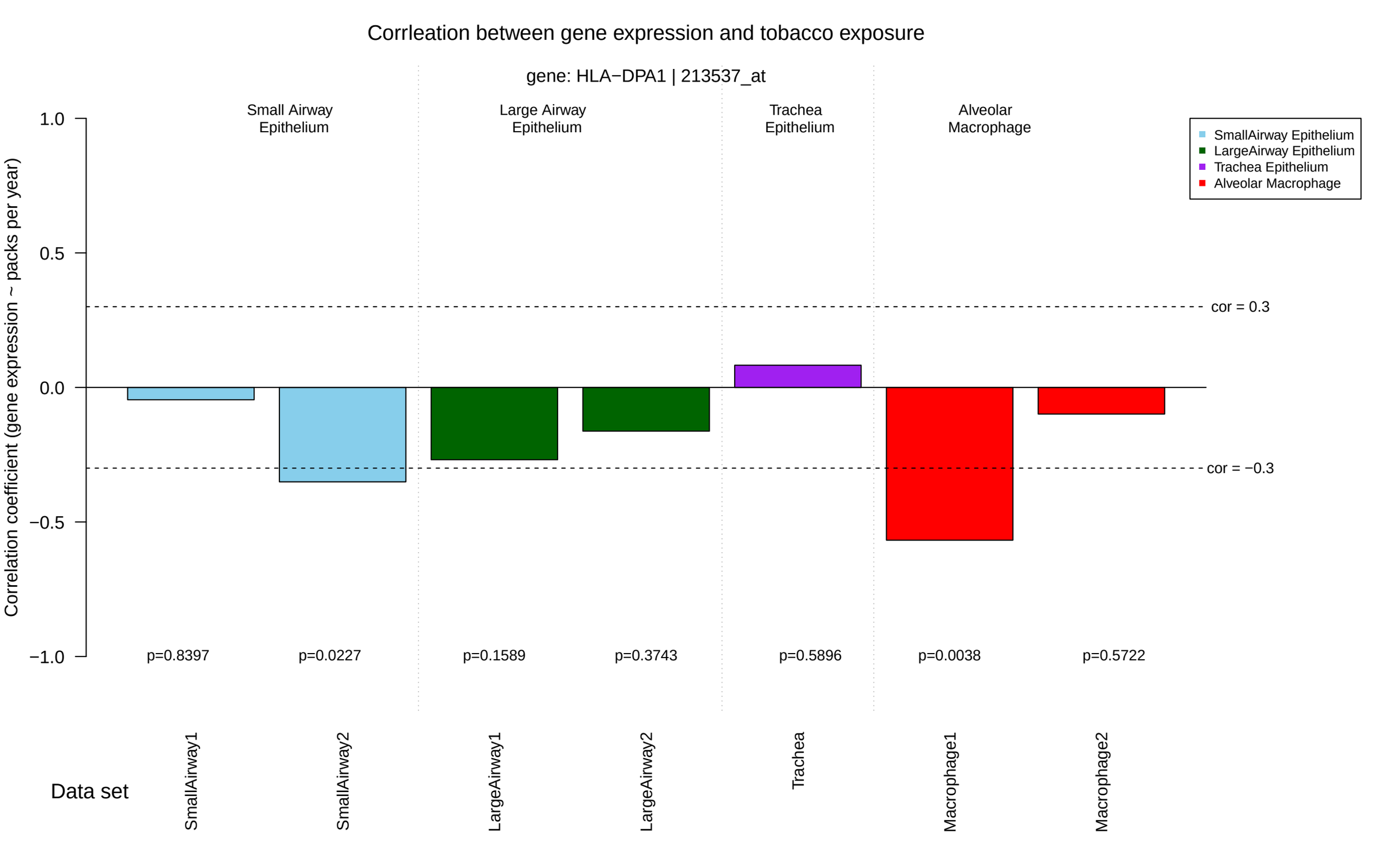

set: LargeAirway1 (p=0.7088)
HLA−DPA1 | 213537_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1827)
HLA−DPA1 | 213537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8848)
HLA−DPA1 | 213537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0106)
HLA−DPA1 | 213537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9632)
HLA−DPA1 | 213537_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DPA1 | 213537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8397
p=0.0227
p=0.1589
p=0.3743
p=0.5896
p=0.0038
p=0.5722
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2